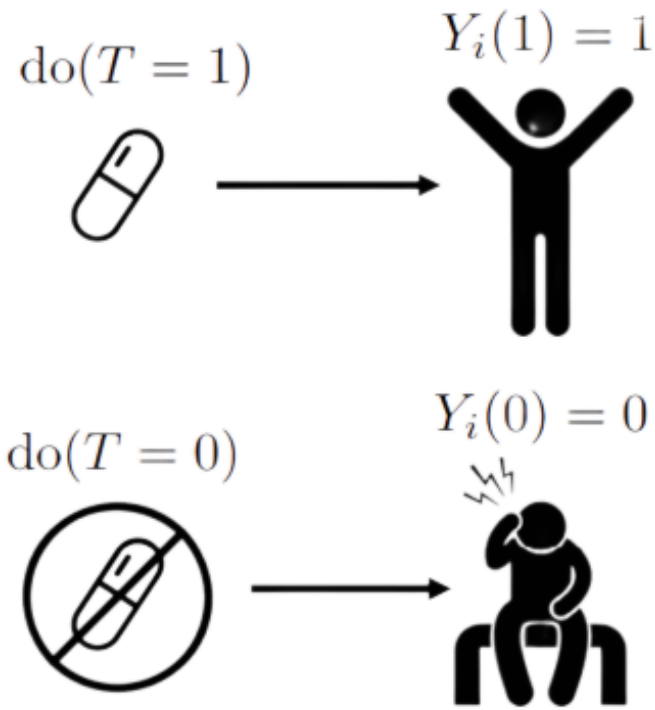
Causal Models

--Counterfactuals

Outline

- Counterfactuals Basics
- Important Application: Mediation
- Representative Works

Fundamental Problem of Causal Inference



T : observed treatment
Y : observed outcome
i : used in subscript to denote a specific unit / individual
$Y_i(1)$: potential outcome under treatment
$Y_i(0)$: potential outcome under no treatment

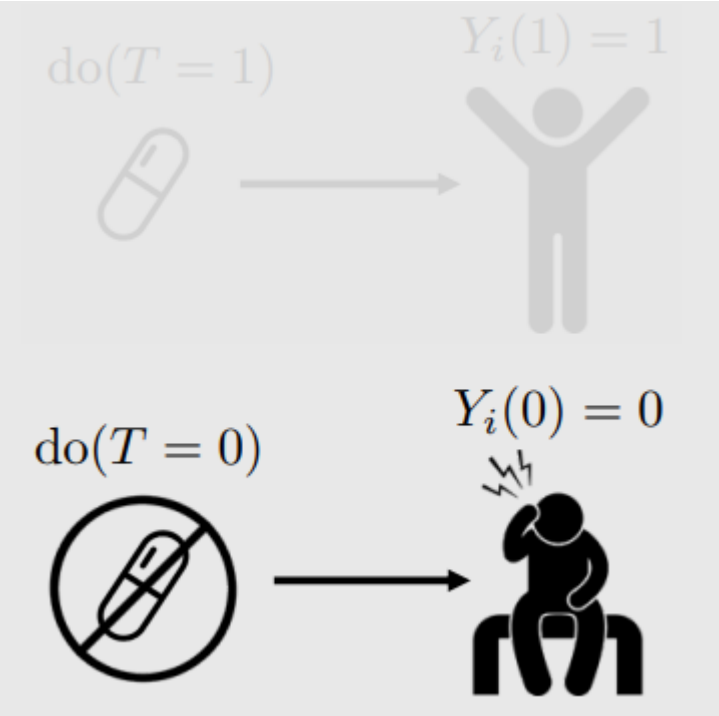
Causal effect

$$Y_i(1) - Y_i(0) = 1$$

Fundamental Problem of Causal Inference

Counterfactual

Factual



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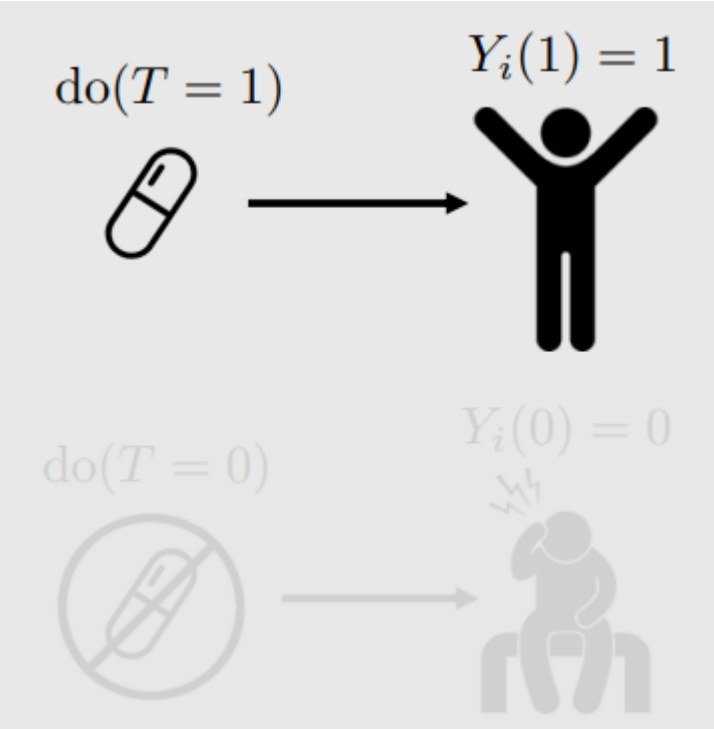

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We can compute counterfactuals using a parametric SCM.

Counterfactuals

Counterfactual: $P(Y(t) \mid T = t', Y = y')$

Counterfactuals

Counterfactual: $P(Y(t) \mid \overbrace{T = t', Y = y'}^{\text{observation}})$

Counterfactuals

observation

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hypothetical condition

Counterfactuals

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Different from CATE: $E[Y(t) \mid X = x]$

Counterfactuals

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Different from CATE: $E[Y(t) \mid X = x] = E[Y \mid do(t), X = x]$

Cannot express counterfactuals using do-notation

Roadmap for Computing Counterfactuals

- Given: Observation of (T,Y) (observation of potential outcome Y(t) where t is the observed value of T)

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- Main ingredient necessary: correct parametric model for the structural equation for Y

- Result: access to counterfactuals $Y(t')$ at the unit-level

Computing Counterfactuals: Simple Example

Y: happy or unhappy (1 or 0)

T: get a dog or don't (1 or 0)

U: unobserved variable describing the individual(1 if dog person; 0 if anti-dog person)

SCM: $T := \ldots$

$Y := UT + (1-U)(1-T)$

Observation: T = 0 and Y = 0 $(Y_u(0) = 0)$

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$$U = 1$$

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Step2: Individualized SCM

$T := \cdots$

$Y := 1 \cdot T + (1-1)(1-T)$

$Y := T$

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$T := \cdots$

$Y := 1 \cdot T + (1-1)(1-T)$

$Y := T$

Step3: $Y_u(1) = 1$

ITE: $Y_u(1) - Y_u(0) = 1$

General Steps for Deterministic Counterfactuals

From Chapter 4 of Pearl et al. (2016)'s Primer:

1. Abduction: Use an observation to determine the value of U

2. Action: Modify the SCM, by replacing the structural equation for T with T := t

3. Prediction: Use the value of U from step 1 and the modified SCM from step 2 to compute the value of Y(t)

Question:

Given the observation T = 1 and Y = 0, compute Y(0) for this individual given the following SCM:

$$T := \cdots$$

$$Y := UT + (1 - U)(1 - T)$$

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Even when we have the structural equation for Y, we can’t always determine counterfactuals with probability 1

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What if we can’t solve for U (function that maps U to Y for a fixed value of T isn’t invertible) ?

Example: Structural equation for Y:

$$Y := \begin{cases} 1 & U = always\ happy \\ 0 & U = never\ happy \\ T & U = dog - needer \\ 1 - T & U = dog - hater \end{cases}$$

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What if we can’t solve for U (function that maps U to Y for a fixed value of T isn’t invertible) ?

Example: Structural equation for Y:

Observation: T = 1 and Y = 0

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Non-Invertible Example

Structural equation for Y:

$$Y := \begin{cases} 1 & U = always\ happy \\ 0 & U = never\ happy \\ T & U = dog - needer \\ 1 - T & U = dog - hater \end{cases}$$

Observation: T = 1 and Y = 0

$$(Y_u(1) = 0)$$

$$(Y_u(0) = ?)$$

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$$Y := \begin{cases} 1 & U = always\ happy \\ 0 & U = never\ happy \\ T & U = dog - needer \\ 1 - T & U = dog - hater \end{cases}$$

$$P(U = always\ happy) = 0.3$$

$$P(U = never\ happy) = 0.2$$

$$P(U = dog - needer) = 0.4$$

$$P(U = dog - hater) = 0.1$$

Observation: T = 1 and Y = 0

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$$P(U = dog - hater) = 0.1$$

Observation: T = 1 and Y = 0

$$P(U = never\ happy \mid T = 1, Y = 0) = \frac{0.2}{0.2+0.1} = \frac{2}{3}$$

$$(Y_u(1) = 0)$$

$$P(U = dog - hater \mid T = 1, Y = 0) = \frac{0.1}{0.2+0.1} = \frac{1}{3}$$

$$(Y_u(0) = ?)$$

$$P(Y_u(0) = 1) = \frac{1}{3}, P(Y_u(0) = 0) = \frac{2}{3}$$

General Steps for Probabilistic Counterfactuals

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No Unit-level Counterfactuals without Parametric Model

Main ingredient necessary for computing counterfactuals: parametric model for the structural equation for Y

Having such a parametric model is a strong assumption.

Without it, we are stuck with the fundamental problem of causal inference.

Question

Given the observation T = 1 and Y = 1, compute Y(0) for this individual given the following SCM and prior:

$$Y := \begin{cases} 1 & U = always\ happy \\ 0 & U = never\ happy \\ T & U = dog - needer \\ 1 - T & U = dog - hater \end{cases}$$

$$P(U = always\ happy) = 0.3$$

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Population-level Doesn't Require a Parametric Model

Population-level counterfactual: $E[Y(t) \mid T = t']$

Just like we were able to identify the ATE $E[Y(1) - Y(0)]$ nonparametrically (using just the causal graph), we can do the same with population-level counterfactual quantities, if they are identifiable

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Same with CATEs: $E[Y(1) - Y(0) \mid X = x]$

See Malinsky et al. (2019)'s potential outcome calculus (generalization of do-calculus) for general identification of counterfactual quantities

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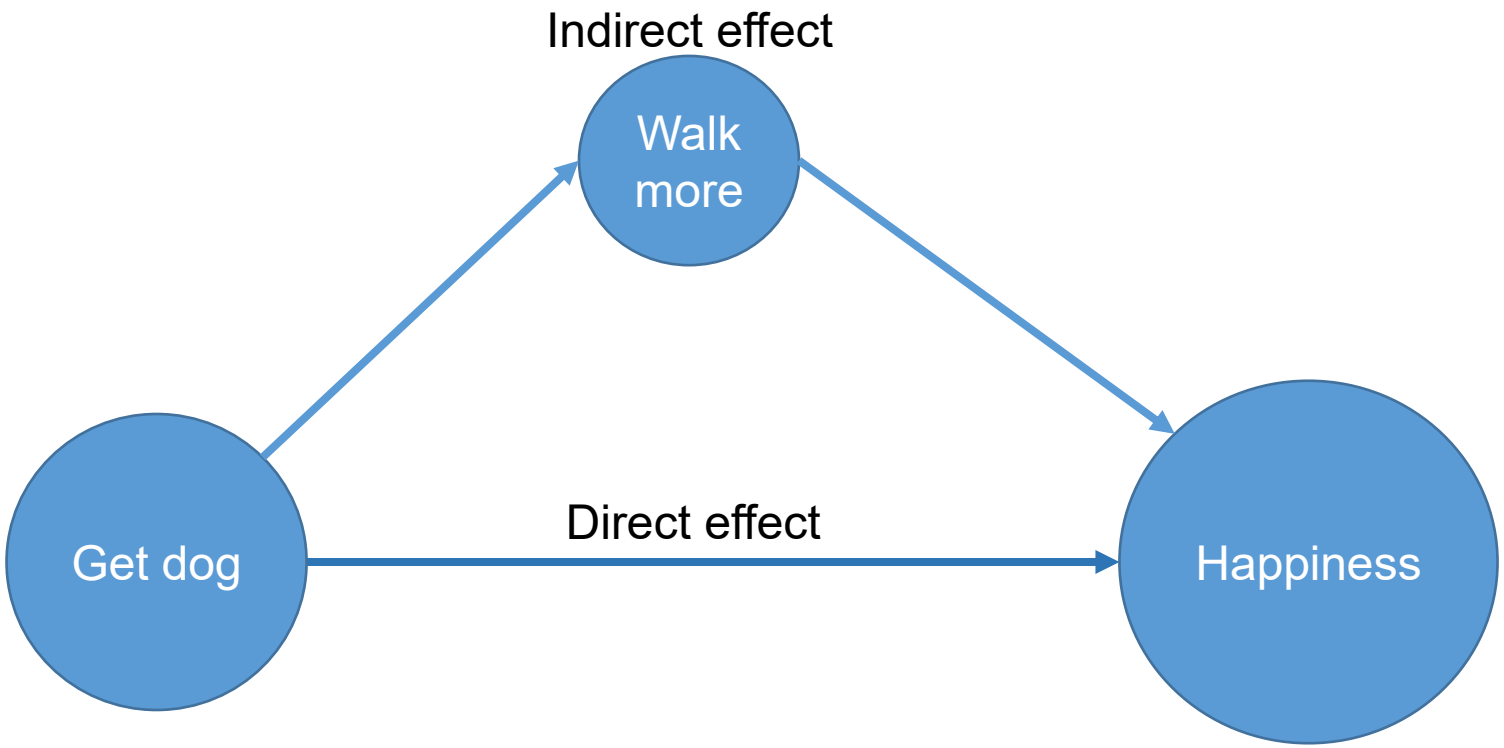
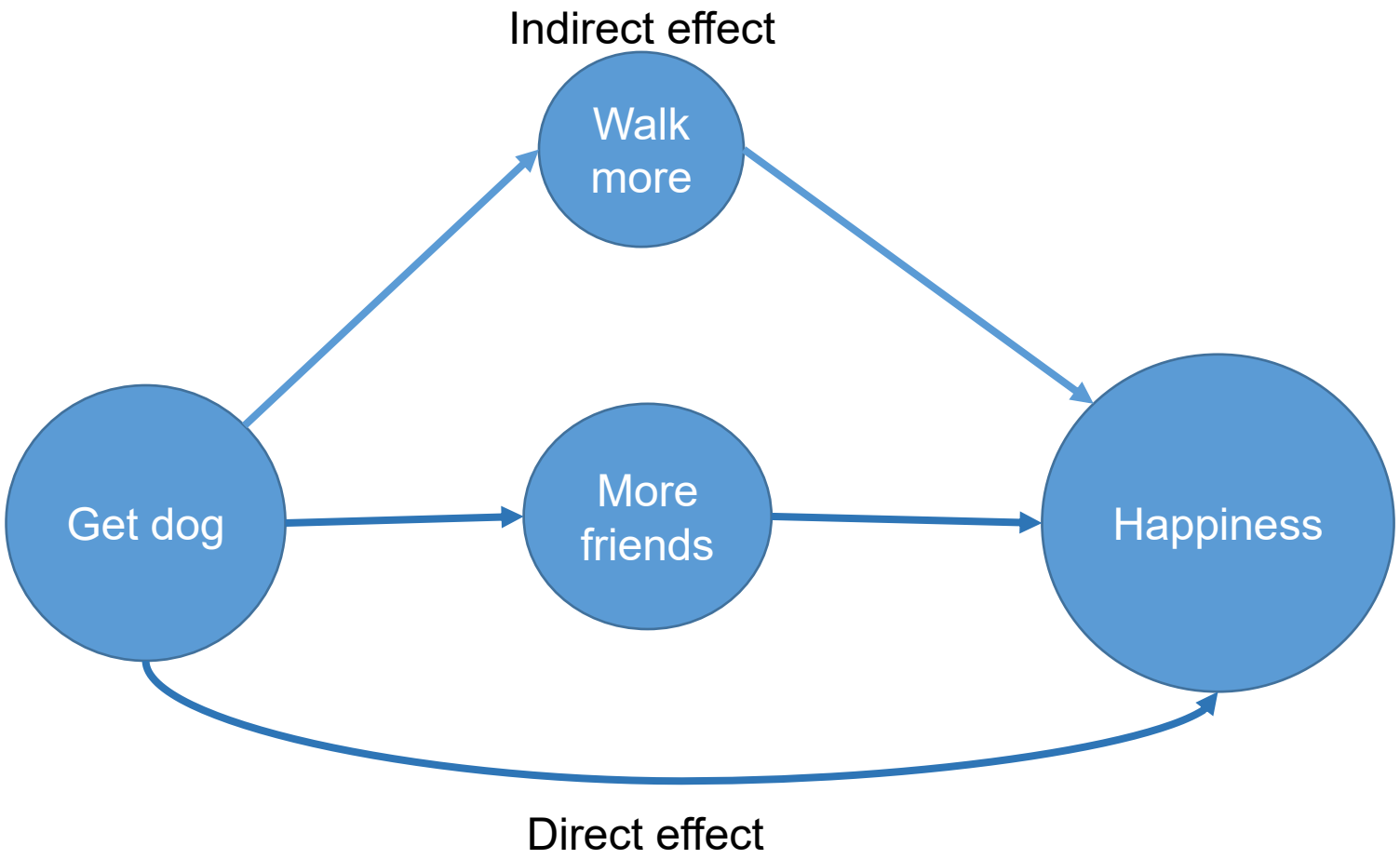
- Important Application: Mediation
- Representative Works

Mediation

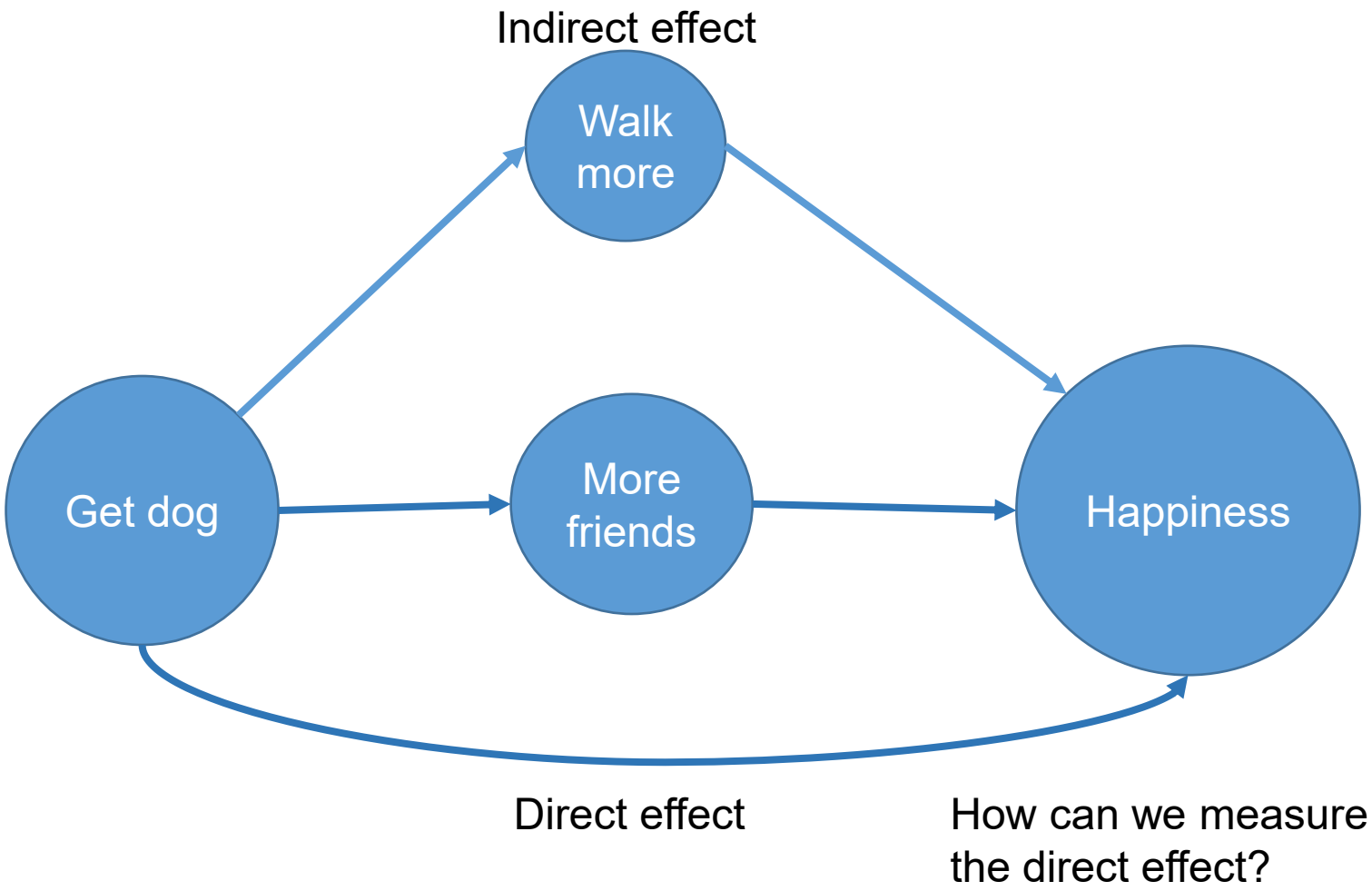
Mediation



Mediation


Indirect effect
Walk more
Get dog
More friends
Happiness
Direct effect

Mediation


Indirect effect
Walk more
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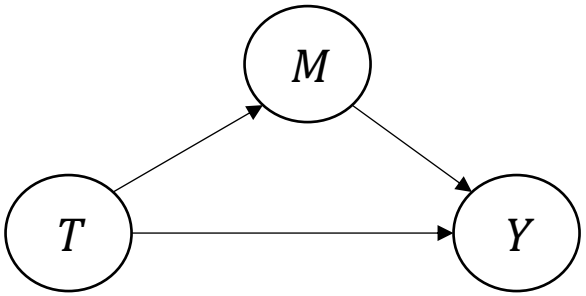
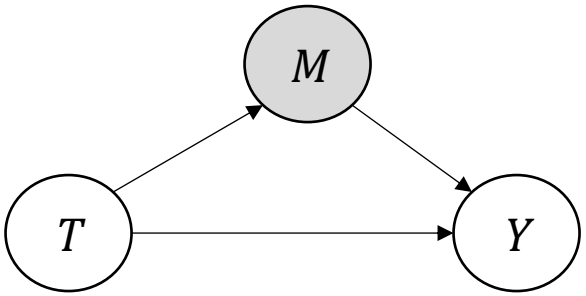
Direct effect
How can we measure the direct effect?

Controlled Direct Effect (CDE)



Controlled Direct Effect (CDE)

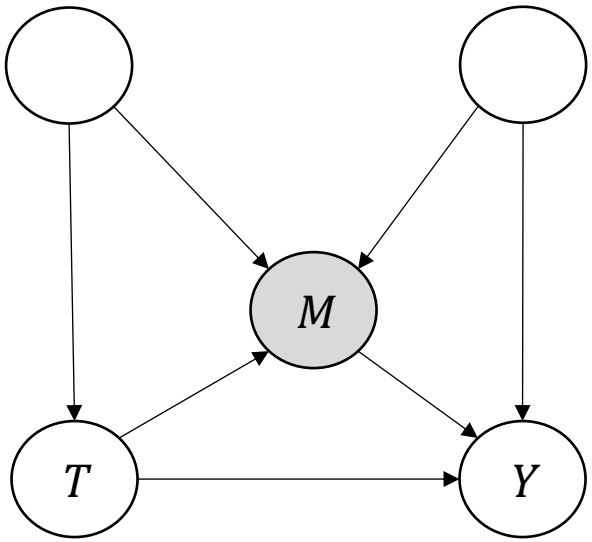
Condition on m



$$E[Y \mid do(T = 1,)\, M = m] - E[Y \mid do(T = 0), M = m]$$

Controlled Direct Effect (CDE)

intervene on m

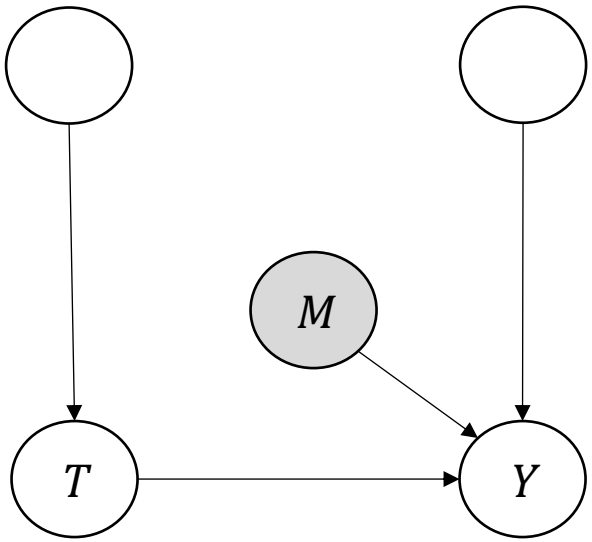


$$\cancel{E[Y \mid do(T = 1,)\, M = m] - E[Y \mid do(T = 0), M = m]}$$

Condition on m could open up other paths

Controlled Direct Effect (CDE)

intervene on m

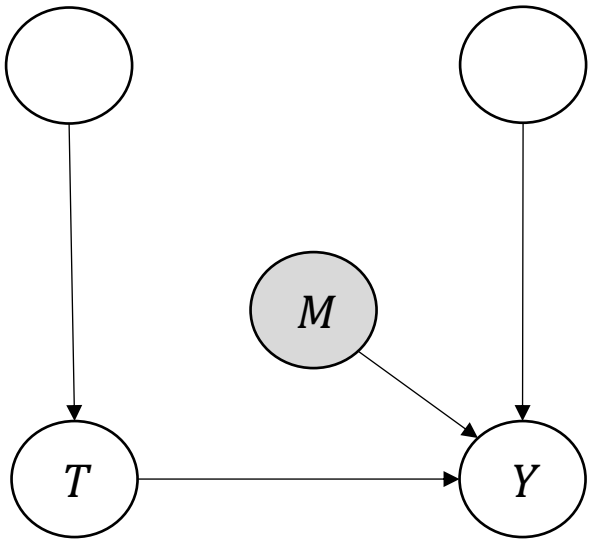


$$\cancel{E[Y \mid do(T = 1,)\, M = m] - E[Y \mid do(T = 0), M = m]}$$

$$E[Y \mid do(T = 1, M = m)] - E[Y \mid do(T = 0, M = m)]$$

We do m then we remove all incoming edges to m
And then we've effectively isolated the flow of causation from t to y in this graph
This is known as controlled direct effect

Controlled Direct Effect (CDE)



Problems:

- CDE is specific to the arbitrary choice of m
- How do we get the indirect effect? Can't just subtract the CDE from the total effect?

$$\cancel{E[Y \mid do(T = 1,)\, M = m] - E[Y \mid do(T = 0), M = m]}$$

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Natural Direct and Indirect Effects

Subscript notation:

$$E[Y_{t,m} \triangleq E[Y \mid do(T = t, M = m)]] \quad E[M_t] \triangleq E[M \mid do(T = t)]$$

$$CDE \triangleq \mathrm{E}[Y_{1,m} - Y_{0,m}]$$

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$$NDE \triangleq \mathrm{E}[Y_{1,M_0} - Y_{0,M_0}]$$

$$NIE \triangleq \mathrm{E}[Y_{0,M_1} - Y_{0,M_0}]$$

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$$TE = NDE - NIE_r$$

Recall problems with CDE:

- CDE is specific to the arbitrary choice of m
- How do we get the indirect effect? Can't just subtract the CDE from the total effect

For example in linear setting, $TE = NDE + NIE$

Question

Show that $TE = NDE - NIE_r$, where $NIE_r \triangleq E[Y_{1,M_0} - Y_{1,M_1}]$.

Comparison of Controlled vs. Natural Mediation

CDE can always be measured via experiments (do-operator), but it has no clear undirect effect since there is no decomposition

NDE cannot always be measured via experiments since it is counterfactual, but it allows for complete decomposition of the total effect into the NDE and NIE, which is what we'd like in mediation analysis

When We Can Measure NDE and NIE

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Adjustment set W

Sufficient conditions for identifying NDE:

- No member of W is a descendant of T
- W blocks all backdoor paths from M to Y

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$$NDE = \sum_{m}\sum_{w}(E[Y \mid do(T=1, M=m), W=w] - E[Y \mid do(T=0, M=m), W=w])$$
$$\times P(M=m|do(T=0), W=w)P(W=w)$$

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How can we get rid of these do operators to identify NDE ?

When We Can Measure NDE and NIE

Adjustment set W

Sufficient conditions for identifying NDE:

- No member of W is a descendant of T
- W blocks all backdoor paths from M to Y
- $P(M = m|do(T = 0), W = w)$ is identifiable (e.g. no unblockable backdoor paths from T to M)
- $E[Y \mid do(T = 1, M = m), W = w]$ is identifiable (e.g. no unblockable backdoor paths from T to Y)

$$NDE = \sum_m \sum_w (E[Y \mid do(T = 1, M = m), W = w] - E[Y \mid do(T = 0, M = m), W = w])$$
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$$\times P(M = m|do(T = 0), W = w)P(W = w)$$

$$= \sum_m \sum_w (E[Y \mid T = 1, M = m, W = w] - E[Y \mid T = 0, M = m, W = w])$$

$$\times P(M = m| T = 0, W = w)P(W = w)$$

Question:

Come up with your own example of mediation and the corresponding graph. Then, determine whether you can identify the NDE and NIE from observation data.

Path-Specific Effects

Measure causal effects along arbitrary path or set of paths in the causal graph

See [“Identifiability of Path-Specific Effects” (Avin et al.,2015)]

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Representative Works

- CV
  - Counterfactual VQA: A Cause-Effect Look at Language Bias (CVPR 2021)
  - Counterfactual Attention Learning for Fine-Grained Visual Categorization and Re-identification (ICCV 2021)
  - Counterfactual zero-shot and open-set visual recognition (CVPR 2021)
  - Counterfactual samples synthesizing for robust visual question answering (CVPR 2020)
  - Cycle-Consistent Counterfactuals by Latent Transformations (CVPR 2022)
  - Towards robust classification model by counterfactual and invariant data generation (CVPR 2021)

Representative Works

- NLP
  - Counterfactual inference for text classification debiasing (ACL 2021)
  - DoCoGen: Domain Counterfactual Generation for Low Resource Domain Adaptation (ACL 2022)
  - Debiasing NLU models via Causal Intervention and Counterfactual Reasoning (AAAI 2022)
  - Counterfactuals to Control Latent Disentangled Text Representations for Style Transfer (ACL 2021)
  - Polyjuice: Generating counterfactuals for explaining, evaluating, and improving models (ACL 2021)

Representative Work1: Counterfactual VQA: A Cause-Effect Look at Language Bias

Motivation: 1) VQA models may rely on spurious linguistic correlations rather than multi-modal reasoning. 2) It will fail to generalize well if they simply memorize the strong language priors in the training data. 3) the goal is to mitigate language bias in VQA

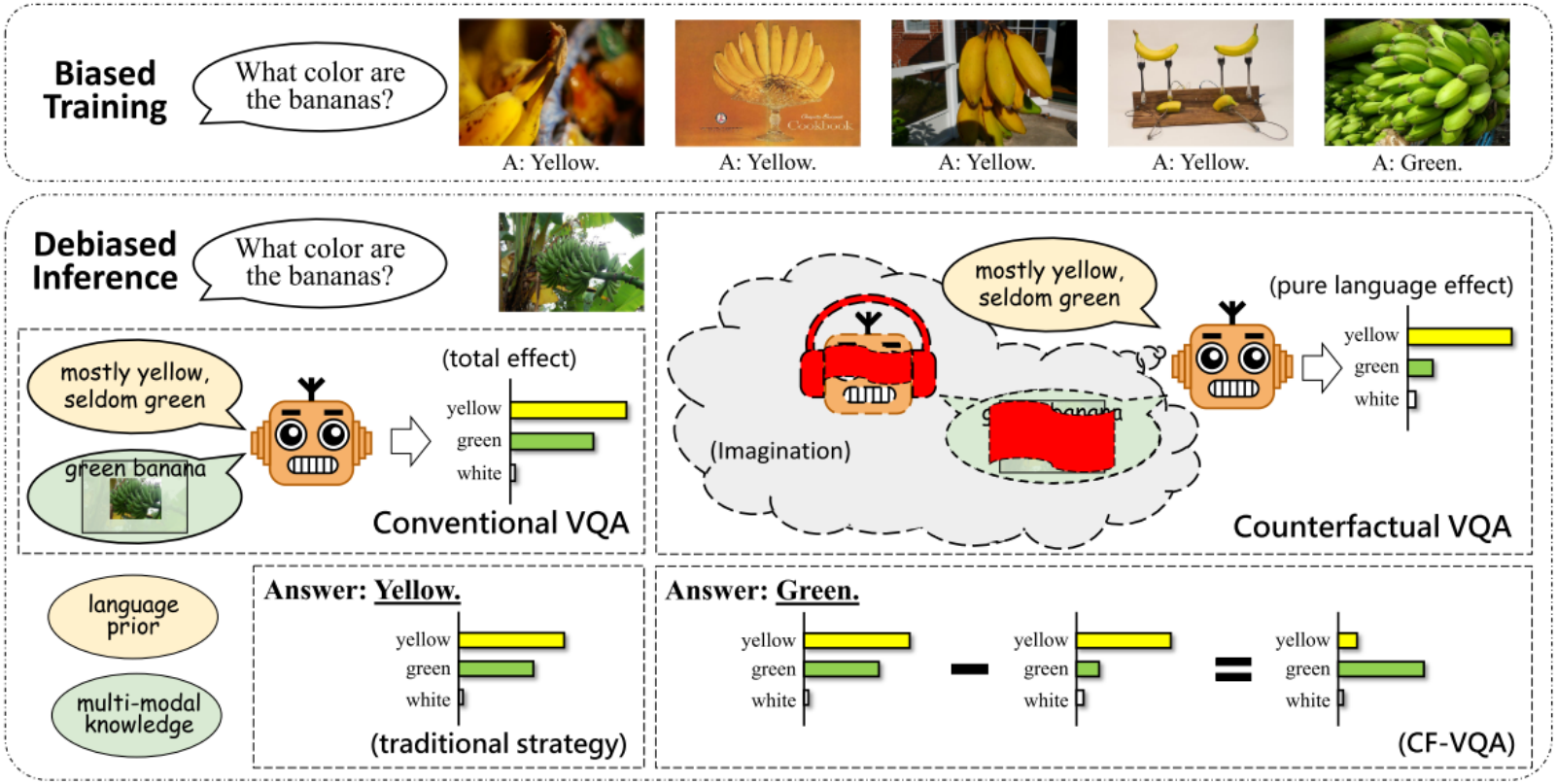


Figure 1: Our cause-effect look at language bias in VQA. Conventional VQA depicts the fact where machine hears the question and extracts the multi-modal knowledge. Counterfactual VQA depicts the scenario where machine hears the question but the knowledge is blocked. We subtract the pure language effect from the total effect for debiased inference.

Representative Work1: Counterfactual VQA: A Cause-Effect Look at Language Bias

SCM and Method: capture the language bias as the direct causal effect of questions on answers and reduce the language bias by subtracting the direct language effect from the total causal effect

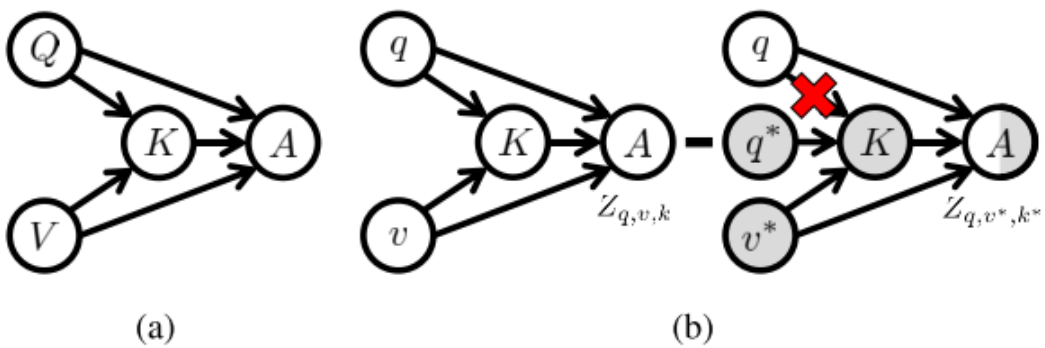


Figure 3: (a) Causal graph for VQA. Q: question. V: image. K: multi-modal knowledge. A: answer. (b) Comparison between conventional VQA (left) and counterfactual VQA (right). White nodes are at the value $V=v$ and $Q=q$ while gray nodes are at the value $V=v^*$ and $Q=q^*$.

$$NDE = Z_{q,v^*,k^*} - Z_{q^*,v^*,k^*}. \quad (8)$$

$$TIE = TE - NDE = Z_{q,v,k} - Z_{q,v^*,k^*}. \quad (9)$$

Representative Work1: Counterfactual VQA: A Cause-Effect Look at Language Bias

Experimental Performance

Table 6: **Comparison on VQA-CP v2 val set**. "Base." indicates the VQA base model.

		VQA-CP v2					VQA v2
		val (in-domain)				test (OOD)	val (in-domain)
	Base.	All	Y/N	Num.	Other	All	All
GRLS [22]	–	56.90	69.23	42.50	49.36	42.33	51.92
GradSup[42]	–	62.4	77.8	43.8	53.6	46.8	–
RandImg [44]	UpDn	54.24	64.22	34.40	50.46	55.37	57.24
CF-VQA (HM)	UpDn	65.47	79.09	45.86	57.86	49.74	63.73
CF-VQA (SUM)	UpDn	60.29	66.32	47.48	57.96	51.27	62.49
CF-VQA (HM)	S-MRL	63.08	75.76	44.88	55.99	53.55	63.54
CF-VQA (SUM)	S-MRL	57.86	66.24	44.98	53.38	55.05	60.94

Representative Work2: Counterfactual Attention Learning for Fine-Grained Visual Categorization and Re-identification

Motivation: there are biases for both attributes and environment, which indicates either background and single part are not reliable clues for classification.

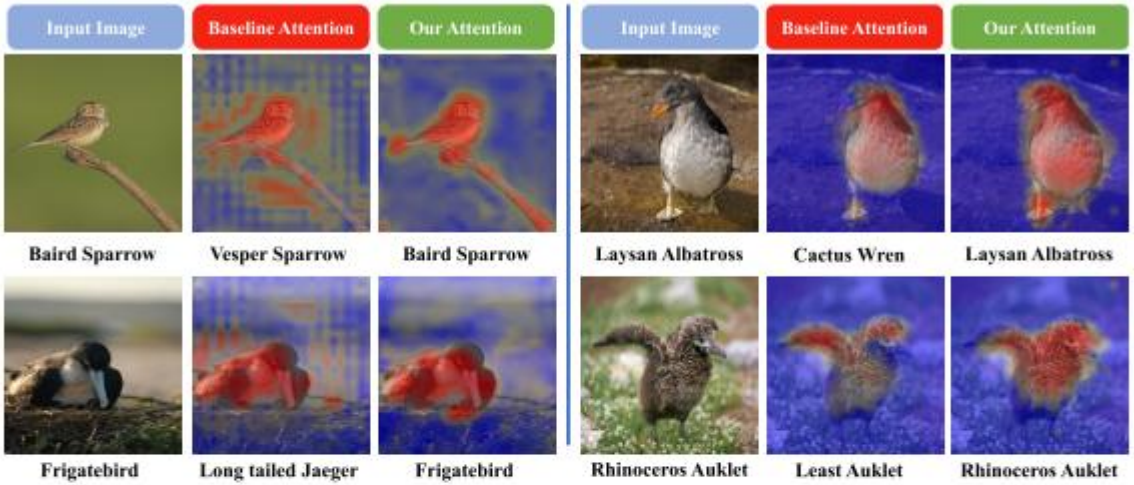
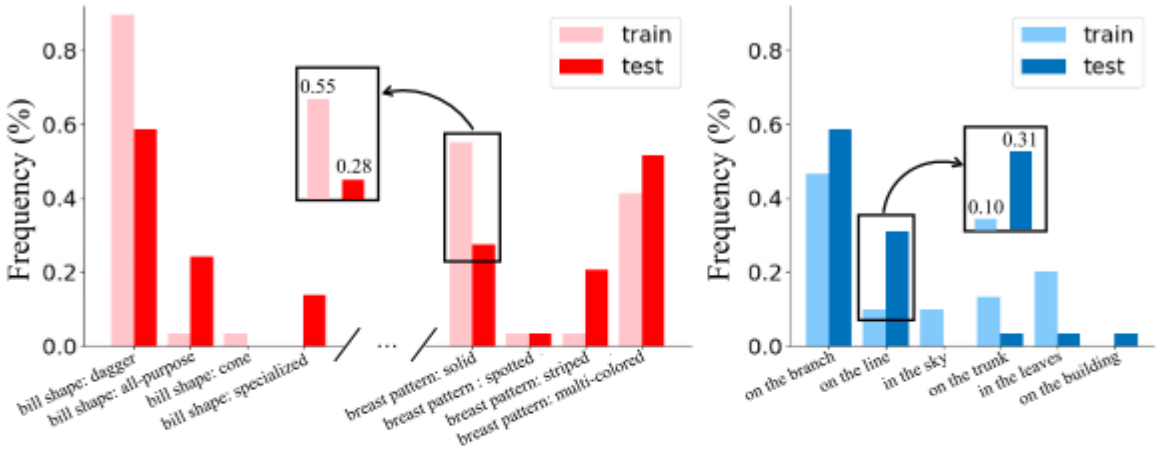


Figure 1: Attention visualization on CUB. We respectively show the original images, baseline attention maps, and attention maps with counterfactual learning. In the left part, we observe that our attention maps can better focus on the object. While comparisons in the right part show that our models prefer to look at the whole object instead of some parts. Best in color.

Representative Work2: Counterfactual Attention Learning for Fine-Grained Visual Categorization and Re-identification

Motivation: biases on the training and testing sets



(a) The attributes (b) The environments

Figure 2: The biases of both intrinsic attributes and external environments on CUB. We demonstrate the biases on the training and testing sets by the statistics the frequencies in different attributes and environments, taking the *Ringed Kingfisher* as an example.

Representative Work2: Counterfactual Attention Learning for Fine-Grained Visual Categorization and Re-identification

SCM and Method: Adopt counterfactual intervention to investigate the effects of the learned visual attention

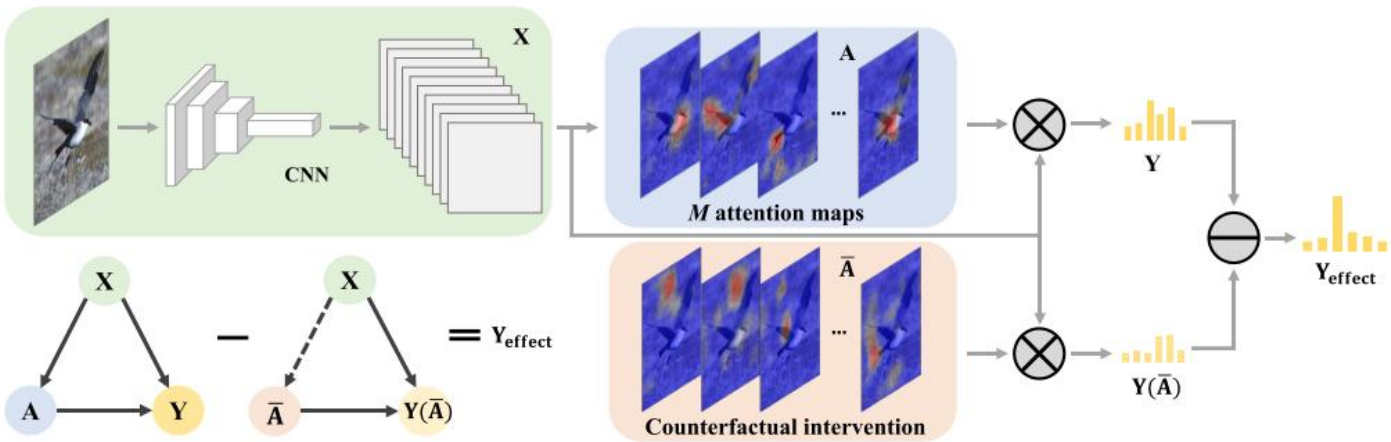
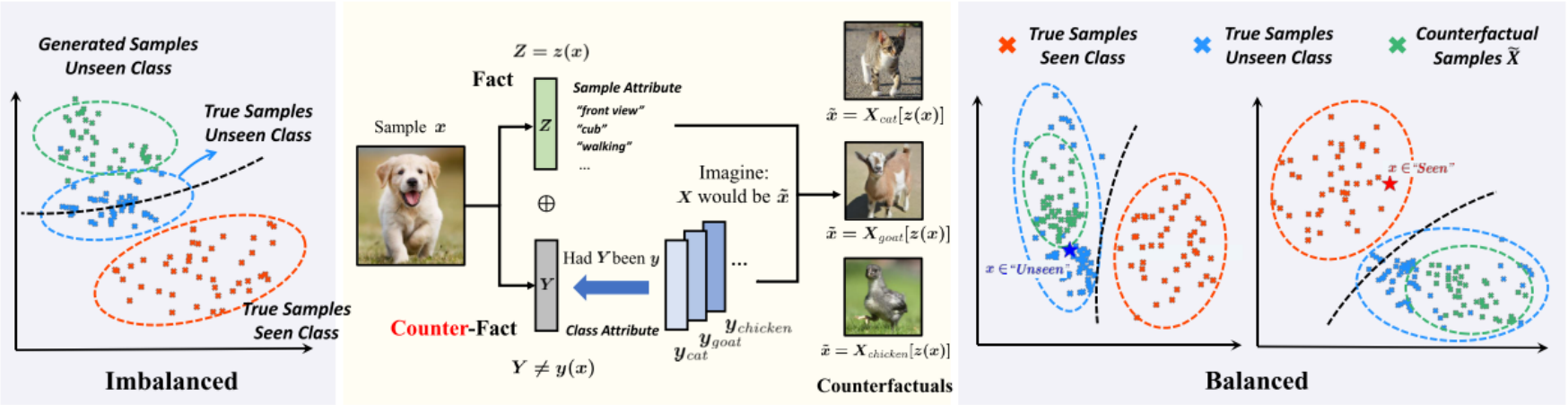


Figure 3: The overall framework of our CAL method. We first apply the counterfactual intervention for original attention by replacing with random attentions. Then, we subtract the counterfactual classification results from original classification to analyze the effects of learned visual attention and maximize them in the training process.

$$Y_{\text{effect}} = \mathbb{E}_{\bar{\mathbf{A}} \sim \gamma}[Y(A=\mathbf{A}, X=\mathbf{X}) - Y(do(A=\bar{\mathbf{A}}), X=\mathbf{X})],$$

Representative Work3: Counterfactual zero-shot and open-set visual recognition

Motivation: **1)** the common challenge of Both Zero-Shot Learning (ZSL) and Open-Set Recognition (OSR) is generalizing to the unseen-classes by only training on the seen-classes. **2)** the generated samples for unseen-classes are often out of the true distribution, which causes severe recognition rate imbalance between the seen-class (high) and unseen-class (low).



(a) (b) (c)

Figure 2: (a) t-SNE [37] plot of the CUB [61] samples in ZSL using a conventional unseen generation method [41], where a single pair of seen- and unseen-class is shown to avoid clutter. Due to the out-of-distribution generation (green), the decision boundary (black dashed lines) is imbalanced between the true seen (red) and unseen (blue) samples. (b) Illustration of our counterfactual generation. (c) t-SNE plot of the CUB using our counterfactual generation. The decision boundary is balanced. See Figure 4 for more examples of OSR.

Representative Work3: Counterfactual zero-shot and open-set visual recognition

Given a sample x, we use the GCM to generate counterfactual samples $\tilde{\mathbf{x}} = X_{\mathbf{y}}[z(\mathbf{x})]$ following the three steps of computing counterfactual :

- **Abduction-**"given the fact that Z = z(x)". We solve for the endogenous sample attribute z(x) given the evidence X = $\tilde{x}$.
- **Action-**"had Y been y" , we intervene on Y by discarding the inferred value and setting Y as y
- **Prediction-**"X would be $\tilde{x}$". Conditioning on the inferred Z = z(x) (fact) and the intervention target Y = y (counter- fact), we can generate the counterfactual sample $\tilde{x}$ from P_{θ}(X|Z = z(x), Y = y)

Representative Work3: Counterfactual zero-shot and open-set visual recognition

We want that the above counterfactual sample lies in the true distribution of the seen or unseen samples.

Definition (Counterfactual Faithfulness). *Given* $\mathbf{x} \in \mathcal{X}$, *the counterfactual generation* $\tilde{\mathbf{x}}$ *using a GCM is faithful whenever* $\tilde{\mathbf{x}} \in \mathcal{X}$.

This allows us to build a binary classifier on seen/unseen by applying the **Consistency Rule**

$$y^*(\mathbf{x}) = \mathbf{y} \Longrightarrow X_{\mathbf{y}}[z(\mathbf{x})] = \mathbf{x}, \qquad (1)$$

$$X_{\mathbf{y}}[z(\mathbf{x})] \neq \mathbf{x} \Longrightarrow y^*(\mathbf{x}) \neq \mathbf{y}. \qquad (2)$$

Representative Work3: Counterfactual zero-shot and open-set visual recognition

Method: apply the **Consistency Rule** to perform unseen/seen binary classification and make the generation is not **Counterfactual Faithful**

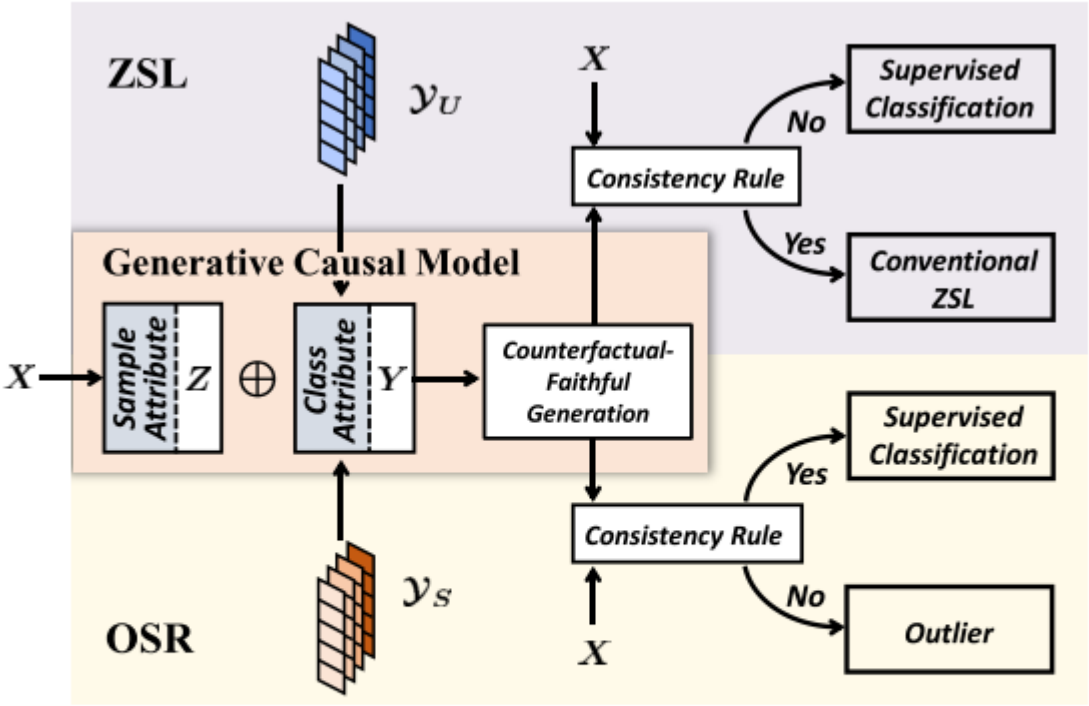


Figure 1: Our counterfactual framework for ZSL and OSR. Here $\oplus$ denotes vector concatenation, $\mathcal{Y}_U$ and $\mathcal{Y}_S$ denote the set of unseen and seen class attributes, respectively.

Representative Work4: Cycle-Consistent Counterfactuals by Latent Transformations

Motivation:

- CounterFactual (CF) visual explanations try to find images similar to the query image that change the decision of a vision system to a specified outcome.
- Existing methods either require inference-time optimization or joint training with a generative adversarial model which makes them time-consuming and difficult to use in practice.

Representative Work4: Cycle-Consistent Counterfactuals by Latent Transformations

Method: **1)** It learns a latent transformation that automatically generates visual CFs by steering in the latent space of generative models. **2)** It uses cycle consistency between the query and CF latent representations which helps training to find better solutions.

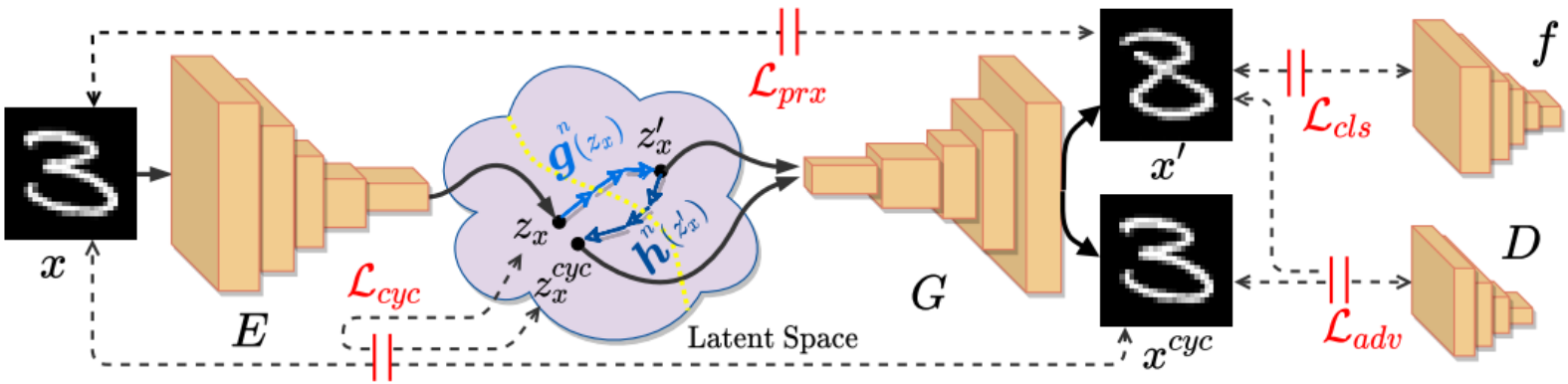


Figure 1. **Cycle-Consistent Counterfactuals by Latent Transformations (C3LT)**. This figure illustrates the architecture of our proposed framework. After a latent code $\boldsymbol{z}_x$ is obtained, our method transformed it to the CF latent code $\boldsymbol{z}'_x$ using g. The CF example can be obtained by $\boldsymbol{x}' = G\left(g^n\left(\boldsymbol{z}_x\right)\right)$. The inputs to the loss functions are outlined via dashed lines. The classifier f and discriminator D are only used during the training. (Best viewed in color)

Representative Work5: Counterfactual inference for text classification debiasing

Motivation: Today's text classifiers inevitably suffer from unintended dataset biases, especially the document-level label bias and word-level keyword bias, which may hurt models' generalization.

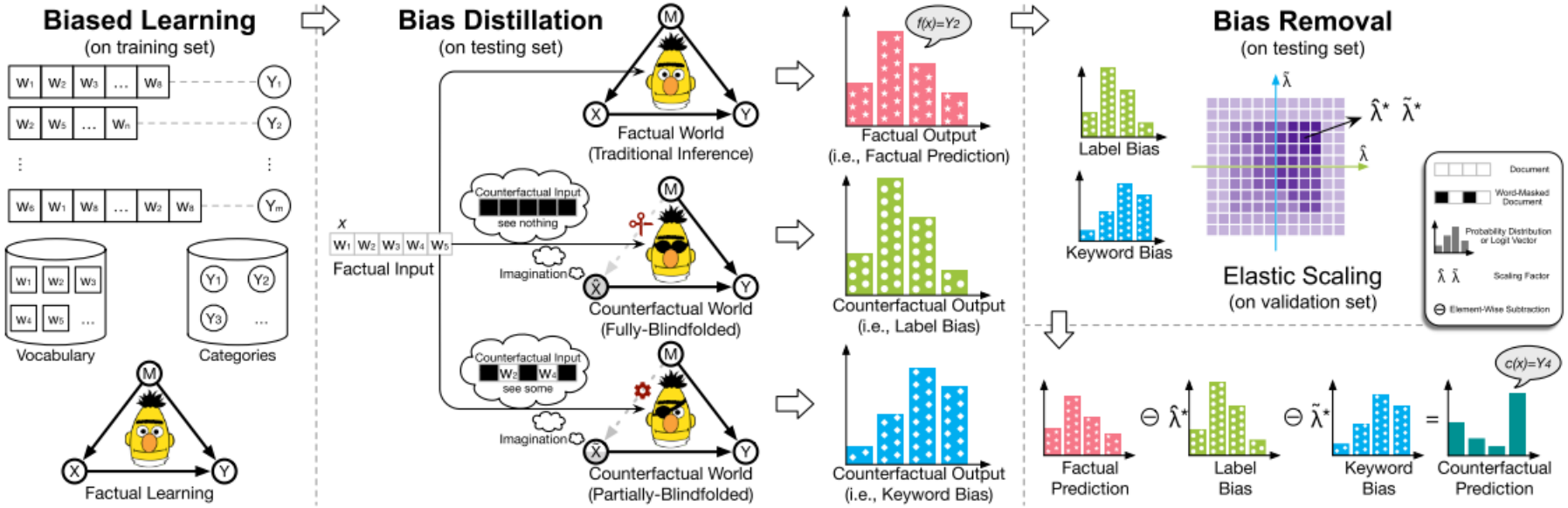
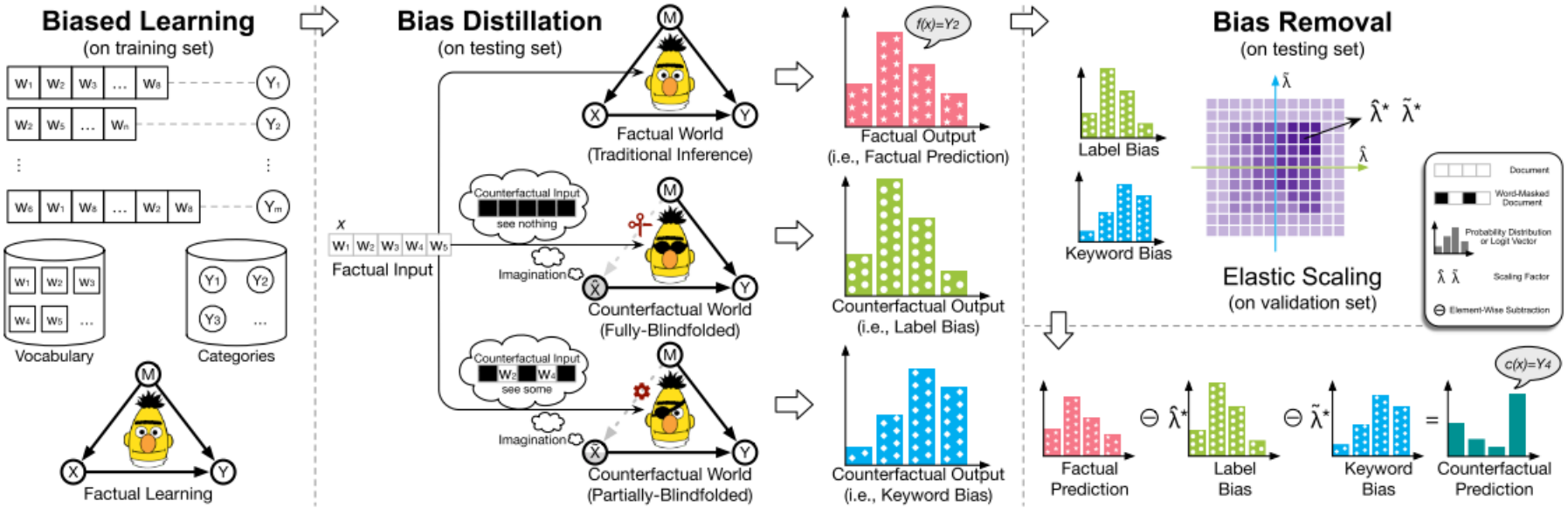


Figure 1: The architecture of our proposed model-agnostic framework (CORSAIR). Specifically, CORSAIR first trains a base model on the training data directly so as to preserve the dataset biases in the trained model. In the inference phase, given a factual input document, CORSAIR first imagines its two types of counterfactual documents to produce two counterfactual outputs as the distilled label bias and keyword bias. Finally, CORSAIR searches two adaptive parameters to perform bias removal to produce a counterfactual prediction for a debiased answer.

Representative Work5: Counterfactual inference for text classification debiasing

SCM and Method: In inference, given a factual input document, CORSAIR imagines its two counterfactual counter-parts to distill and mitigate the two biases captured by the poisonous model.



$$c(x) = f(x) \backslash f(\hat{x}) \backslash f(\tilde{x}) = f(x) - \hat{\lambda} f(\hat{x}) - \tilde{\lambda} f(\tilde{x}) \quad (6)$$

Similar to representative work 1

Representative Work6: DoCoGen: Domain Counterfactual Generation for Low Resource Domain Adaptation

Motivation: Natural language processing (NLP) algorithms have become very successful, but they still struggle when applied to out-of-distribution examples.

Original, **Kitchen**: A good knife but Quality Control was poor. The knife is solid and very comfortable in hand, however, when I got it new, the blade is slightly bent. I expect it to be in almost perfect condition, but it's not.	
DoCoGen, **Kitchen → Electronics**: A good product but Quality Control was poor. The ipod is very easy to use and very comfortable in hand, however, when I got it new, the ipod is slightly flimsy. I expect it to be in almost perfect shape, but it's not.	
Original, **DVD**: The direction of this film is excellent. I love all the characters and the way they interact. The storyline is very important also. It's about religious beliefs and neighbors that interact with each other. It's a well-paced and interesting story that's not like anything else I've ever seen.	
DoCoGen, **DVD → Airline**: The service on this flight is excellent. I love the staff and the way they interact. The safety is very important also. It's nice to have staff and neighbors that can help each other. It's a well-groomed and professional crew that's not like anything else I've ever experienced.	
Original, **Electronics**: That relay board is only good for switching AC loads of 100V or more. If you have a lower voltage load, it's not going to work. For low voltage loads use transistors, MOSFETs or a ULN2803 driver board.	
DoCoGen, **Electronics → Statistics**: That model is only good for data of n or more. If you have a lower n, it's not going to work. For lower n regression use a linear, logistic or a t-test.	

Table 1: Domain-counterfactual textual examples (D-CONs) generated by DoCoGen. Red terms are replaced with green terms through the process of D-CON generation. For additional examples see §A.

Representative Work6: DoCoGen: Domain Counterfactual Generation for Low Resource Domain Adaptation

Method: Given an input text example, the algorithm generates a domain-counterfactual textual example (D-CON) – that is similar to the original in all aspects, including the task label, but its domain is changed to a desired one

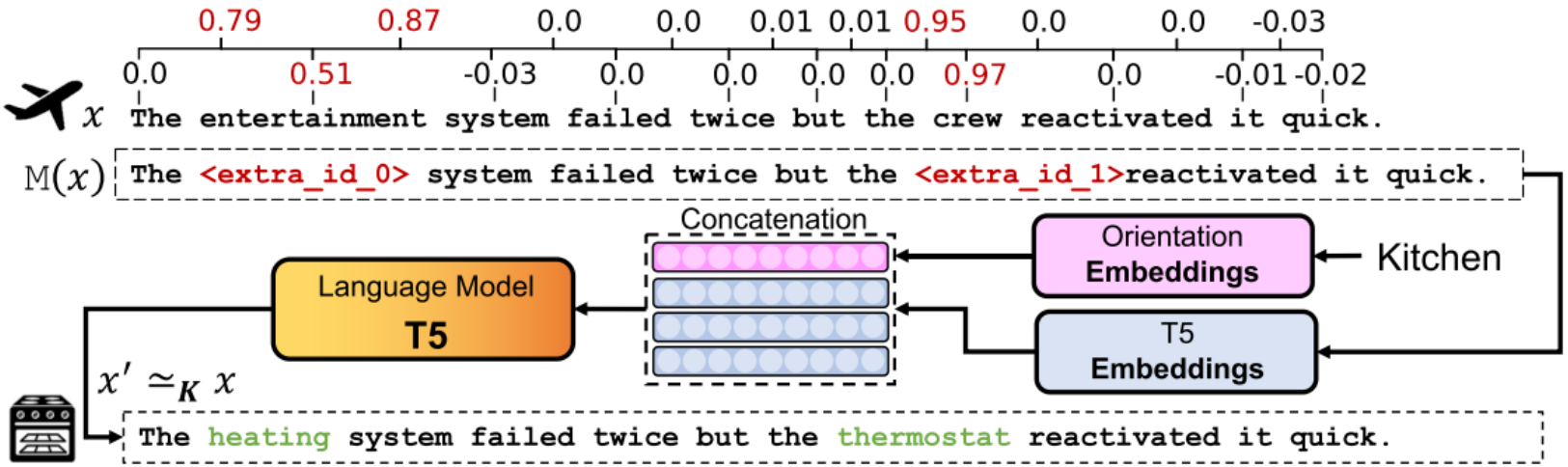


Figure 1: The DoCoGen model. Given a review x from the *airline* domain, we aim to generate a D-CON from the *kitchen* domain. We first corrupt the domain of the example by masking domain specific terms. The numbers above the input words are the masking scores of uni-grams and bi-grams. Terms with scores above a threshold ($\tau = 0.08$) are masked. In the reconstruction step we use a T5-based generation model to generate the D-CON $x' \simeq_{\mathbf{K}} x$. The input of the model is a concatenation of the *orientation vector* that represents the target domain with the model's embedding vectors which correspond to the tokens of the masked example $\mathrm{M}(x)$.

Similar to representative work3

Representative Work7: Debiasing NLU models via Causal Intervention and Counterfactual Reasoning

Motivation: Natural Language Understanding (NLU) models are prone to relying on annotation biases of the datasets as a shortcut, which goes against the underlying mechanisms of the task of interest

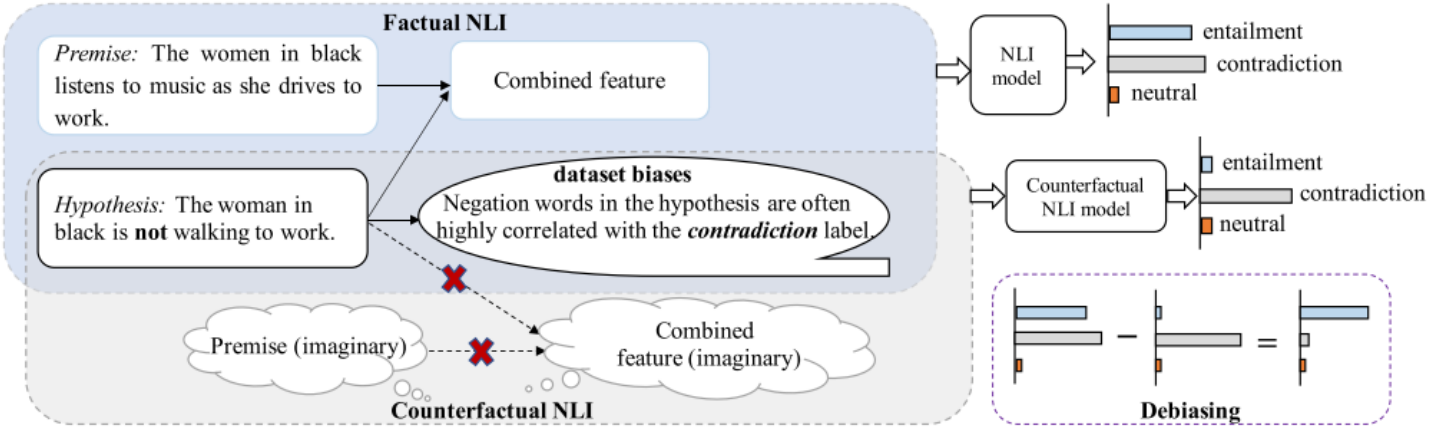
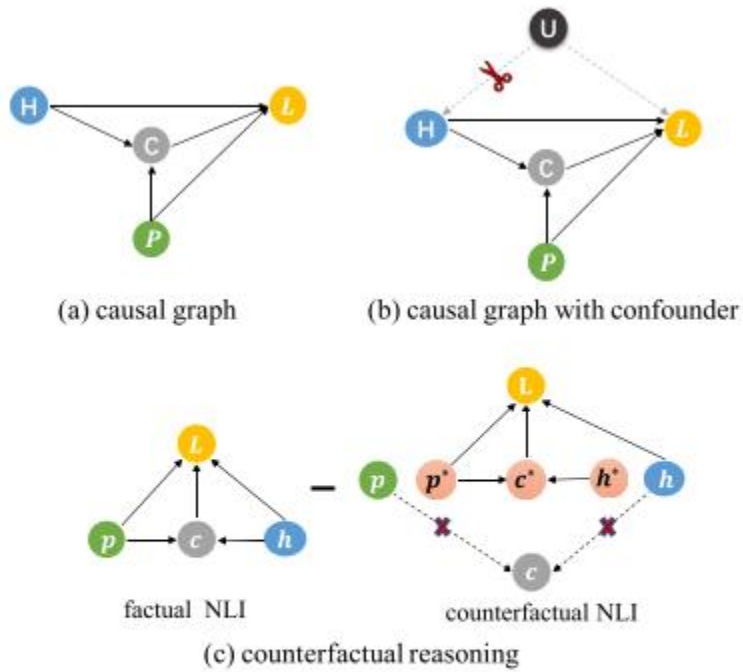


Figure 1: An illustration of factual and counterfactual NLI, as well as the debiasing strategy. Factual NLI depicts the fact where model sees the hypothesis and extracts the combined feature of premise and hypothesis. Counterfactual NLI means that model sees the hypothesis but the combined feature and premise are coming from the imagined world.

Representative Work7: Debiasing NLU models via Causal Intervention and Counterfactual Reasoning

SCM and Method: **1)** To remove confounder, the backdoor adjustment with causal intervention is utilized to find the true causal effect during training **2)** In inference process, we formulate the bias as the direct causal effect and remove it by pursuing the indirect causal effect with counterfactual reasoning.



$$NDE = S_{h,p^*,c^*} - S_{h^*,p^*,c^*}$$

$$TIE = TE - NDE = S_{h,p,c} - S_{h,p^*,c^*}$$

Figure 4: Causal graph of NLI and the comparison between factual NLI and counterfactual NLI.

Summary

- Counterfactuals Basics
  - Roadmap for Computing Deterministic Counterfactuals
  - General Steps for Probabilistic Counterfactuals
- Important Application: Mediation
  - CDE、NDE、NIE
  - When We Can Measure NDE and NIE
- Representative Works

Causal Inference

- 参考：https://www.bradyneal.com/causal-inference-course
- https://www.inference.vc/causal-inference-3-counterfactuals/